



EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2027	4YRS B.S	IIT Kharagpur	8.43 / 10
2023	Central Board of Secondary Education (CBSE)	Kendriya Vidyalaya IIT Kharagpur	91.2%
2021	Central Board of Secondary Education (CBSE)	Kendriya Vidyalaya IIT Kharagpur	97.6%

INTERNSHIPS

Data Science Intern | Polestar Analytics

May 2026 - Jul 2026

Project: Agentic Stock Manager | Human-in-the-loop system for automated supply-chain replenishment

- Built a **multi-agent system** to detect stock threshold breaches and generate **purchase orders** from live inventory, warehouse, and product data.
- Integrated real-time APIs, multi-LLM routing, purchase order approvals, guardrails, and audit logs into a **6-agent LangGraph workflow**.
- Consolidated per-SKU orders into **one purchase order per vendor** using vendor based batching, supporting auto issue, draft, batch, and suppress

Project: Promotional Uplift Modeling | Time-aware forecasting and causal promotion targeting

- Built a time aware demand forecasting pipeline integrating multi-source retail data with lag, rolling, trend, and **Fourier features**.
- Developed and benchmarked **CatBoost T-, S-, and X-learners** alongside hierarchical Bayesian uplift models using **Qini, AUUC, uplift@K**
- Converted uplift estimates into a capacity constrained **promotion policy**, prioritizing product-weeks by predicted incremental demand

Analyst Programmer Intern | Xceedance Consulting Pvt. Ltd.

May 2025 - Sept 2025

- Architected an end to end insurance document pipeline using **Tesseract OCR** for text extraction and **BERT** based NER for entity extraction
- Fine tuned **BERT** to extract insurance entities from OCR text, achieving **94.2%** accuracy and **0.91** weighted **F1**, reducing costly LLM API calls.
- Implemented **RapidFuzz** based entity **normalization** to map noisy NER outputs to standardized reference records, reducing manual review
- Built a **Streamlit** interface for NER and normalization, integrating **MLflow** tracking and automated **Databricks** ML pipeline execution.

PROJECTS

Fan-Out MCP: Multi-Agent Delegation Plugin | Self Project

- Architected a cross-harness **MCP** runtime for **Claude Code**, Codex and Copilot CLI, dispatching 5-30 subtasks to local and hosted SLMs.
- Implemented an async execution engine with bounded **concurrency**, timeouts, exponential backoff, and cost ceilings for reliable execution.
- Engineered **multi-strategy** aggregation using majority voting, LLM-as-Judge scoring, and Map-Reduce with prompt injection isolation.
- Designed a local-first, default-deny provider framework with pluggable routing, **OpenTelemetry** tracing, and JSONL token accounting.

DhruvGPT - Real-Time Streaming LLM Chat Application | Self Project

- Built an asynchronous **FastAPI** backend with **Server-Sent Events (SSE)** for real-time streaming of NVIDIA LLM responses.
- Designed conversation persistence using **PostgreSQL**, **SQLAlchemy ORM**, and **Alembic** for version-controlled schema migrations.
- Engineered resilient NVIDIA inference integration with streaming tokens, retry logic, rate-limit handling, cancellation, and error recovery.
- Built and evaluated machine learning models for **Kaggle competitions**, using data preprocessing and feature engineering.

LiteHNSW In-Memory Vector Search Engine | Self Project

- Built a C++ **HNSW** search engine from scratch with diversity-aware graph construction and configurable nearest-neighbor search.
- Optimized vector search with **AVX2/AVX-512** distance kernels, aligned memory allocation, and compact graph storage.
- Enabled concurrent queries and exclusive insertion using a shared **mutex** and per-call scratch buffers for linearizable visibility.
- Built **pybind11** bindings for NumPy workflows; benchmarked recall, latency, and memory against **FAISS** using ANN-Benchmarks.

COMPETITION/CONFERENCE

- Codeforces **Expert** (peak 1606; best rank 141 in Div. 2) and solved 1000+ DSA problems across LeetCode, Codeforces, CSES, and CodeChef.
- Ranked top **50** on the final leaderboard and top 10 on the rolling leaderboard in IIT Pokerbots. Title Sponsor: Jane Street, Organizer: IITM
- Secured **4th** place in **Open IIT Data Analytics 2024** by building a supervised learning model with feature engineering to predict flight delays.

SKILLS AND EXPERTISE

Programming Languages: *Proficient:* C/C++, Python, SQL

Familiar: JavaScript, Bash, HTML, CSS

Frameworks and Libraries: C++ STL, FastAPI, AsyncIO, SQLAlchemy, Alembic, Pydantic, pybind11, NumPy

Tools and Platforms: Git, GitHub, Docker, CMake, PostgreSQL, Linux, OpenTelemetry, VS Code, Vercel

Core Skills: DSA, OOP, Database Mgmt, API Development, Async Programming, Concurrency, Memory Mgmt, Performance Optimization

COURSEWORK INFORMATION

Computer Science: Information Retrieval, Machine Learning, Algorithms (Theory & Laboratory) , Programming & Data Structures (Theory & Laboratory) , Operating Systems, Database Management Systems, Deep Learning, Computer Networks, Natural Language Processing

Maths: Linear Algebra & Numerical Analysis, Advanced Calculus, Probability & Statistics, Partial Differential Equations, Transform Calculus

Core: Methods of Prospecting, Pattern Recognition, Borehole Geophysics, Geophysical Signal Processing, Geophysical Inverse & Field Theory

EXTRA CURRICULAR ACTIVITIES

- Promoted to Head of Design at Click KGP, IIT Kharagpur's Photography Club, and later served as Advisor.
- Selected for the Inter IIT Badminton Contingent; captained Nehru Halls badminton team in General Championship Sports.
- Achieved peak ratings of 1500+ in Chess.com Rapid, 1600+ in Blitz, and 1600+ in Matiks Sprint Duels.